H^s DIAGNOSTIC CODE REFERENCE

Complete Reference for the Higgins Decomposition Pipeline Code System

76 Diagnostic Codes · 10 Structural Modes · 5 Severity Levels · 13 Pipeline Stages

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1. Code Format

Every diagnostic code follows a fixed three-part format:

SS-CCC-LLL

Part	Name	Description
SS	**Stage**	2-character pipeline stage identifier (GD, S4–SC, XU, XC, RP)
CCC	**Condition**	3-character condition code identifying what was detected at this stage
LLL	**Level**	3-character severity/information class: INF, WRN, ERR, DIS, or CAL

Example: **S8-NAT-INF** — Stage S8 (Transcendental Squeeze), Condition NAT (NATURAL classification), Level INF (informational).

2. Diagnostic Levels

The five diagnostic levels classify what kind of information each code conveys and how the analyst should respond:

INF — Information

Normal pipeline operation. The step completed and its output is recorded. INF codes form the structural skeleton of every report — they confirm that each step ran and what it measured. A pipeline run with only INF codes means the system is compositionally clean and well-behaved.

WRN — Warning

Attention needed. A measurement fell outside the expected range or a quality threshold was not met. Warnings do not stop the pipeline but indicate that the result should be interpreted with caution. Common triggers: low HVLD R^2, INVESTIGATE classification, EITT failure, extreme carrier hierarchy.

ERR — Error

Input rejected or operation failed. The pipeline cannot produce a valid result from this data. All ERR codes are emitted by the Guard stage and terminate the pipeline before any compositional operations begin. The data must be cleaned and re-submitted.

DIS — Discovery

Noteworthy finding. The pipeline has detected a compositional structure that warrants investigation. Discoveries are the scientific output of the pipeline — they flag Euler-family resonances, carrier dominance, regime transitions, stable ratios, zero crossings, and other structural features that reveal the physics of the system.

CAL — Calibration

Instrument calibration signal. The pipeline's own measurement precision is being quantified, not the data's properties. CAL codes fire when match density is below the natural baseline (indicating the transcendental library may be sparse for this domain) or when transfer entropy is zero (indicating the system has no directed information flow at this aggregation level). They calibrate the instrument, not the subject.

3. Pipeline Stages and Codes

This section documents every diagnostic code organised by the pipeline stage that emits it. For each code: the identifier, short name, diagnostic level, trigger condition (from generate_codes()), and full description with analytical significance.

3.1 GD — Guard (Input Validation)

Pre-pipeline: Validates raw input before any compositional operations. Rejects data with insufficient dimensions, too few observations, NaN/Inf contamination, or extreme scale disparity. All four guards must pass for the pipeline to proceed.

Code	Level	Description & Trigger
GD-D2M-ERR	ERR	**D < 2** Input rejected: D must be >= 2. The simplex requires at least 2 compositional parts. ***Trigger:*** *Emitted by the guard check when the input matrix has fewer than 2 columns (compositional parts).*
GD-N5M-ERR	ERR	**N < 5** Input rejected: N must be >= 5. The variance trajectory needs at least 3 non-zero points for a meaningful HVLD fit. ***Trigger:*** *Emitted when the input matrix has fewer than 5 rows (observations).*
GD-NAN-ERR	ERR	**NaN detected** Input rejected: data contains NaN values. Clean or impute before decomposition. ***Trigger:*** *Emitted when any element of the input matrix is NaN.*
GD-INF-ERR	ERR	**Inf detected** Input rejected: data contains Inf values. Check for division by zero or overflow. ***Trigger:*** *Emitted when any element of the input matrix is Inf or -Inf.*
GD-SCL-ERR	ERR	**Scale disparity** Input rejected: max/min ratio exceeds 1e15. Normalise or partition before decomposition. ***Trigger:*** *Emitted when the ratio of the largest to smallest positive value in the input exceeds 10^{15}.*
GD-ALL-INF	INF	**All guards pass** All 4 input guards passed. Data is valid for H^s pipeline processing. ***Trigger:*** *Always emitted at pipeline start if no guard errors were raised. Confirms data validity.*

3.2 S4 — Simplex Closure

Step 4: Closes raw data onto the D-simplex so that each observation sums to 1.0. Applies multiplicative zero replacement (ZERO_DELTA = 1e-6) if any component is exactly zero, preventing log-domain singularities in subsequent steps.

Code	Level	Description & Trigger
S4-CLO-INF	INF	**Closure OK** Simplex closure successful. All rows sum to 1.0 within tolerance 1e-12. ***Trigger:*** *Always emitted after closure. Value: the closure check result from step3_closure_check.*
S4-ZRP-INF	INF	**Zero replacement** Multiplicative zero replacement applied. ZERO_DELTA = 1e-6. ***Trigger:*** *Emitted when step3_zero_replacement_applied is True — at least one zero was found and replaced.*
S4-ZRN-INF	INF	**No zeros**

		No zero replacement needed. All input values are positive. ***Trigger:*** *Emitted when step3_zero_replacement_applied is False — all values were already positive.*

3.3 S5 — CLR Transform

Step 5: Applies the Centred Log-Ratio transform: clr(x) = log(x) − (1/D)Σ log(x_j). Maps compositional data from the simplex into Euclidean space where standard linear operations are valid. This is an exact algebraic mapping with no approximation.

Code	Level	Description & Trigger
S5-CLR-INF	INF	**CLR complete** Centred log-ratio transform applied. All rows sum to zero in CLR space. ***Trigger:*** *Always emitted after CLR transform completes. Unconditional.*

3.4 S6 — Aitchison Variance

Step 6: Computes the cumulative Aitchison variance trajectory σ^2_A(t) across all time indices. This trajectory captures how compositional variability accumulates and is the primary input to the HVLD vertex lock in the next step.

Code	Level	Description & Trigger
S6-VAR-INF	INF	**Variance computed** Cumulative Aitchison variance trajectory σ^2_A(t) computed for all time indices. ***Trigger:*** *Always emitted. Value: step5_sigma2_final (the final cumulative variance).*
S6-ZRO-WRN	WRN	**Zero variance** σ^2_A is zero or near-zero. Data may be constant or degenerate. ***Trigger:*** *Emitted when step5_sigma2_final < 1e-10. Indicates the data has essentially no compositional variability.*
S6-RNG-INF	INF	**Variance range** Variance trajectory range and final value recorded. ***Trigger:*** *Always emitted. Value: step5_sigma2_range (min and max of the variance trajectory).*

3.5 S7 — HVLD Vertex Lock

Step 7: Fits a quadratic polynomial to the variance trajectory and classifies its shape as bowl (a > 0, variance accelerating) or hill (a < 0, variance decelerating). The vertex location is the 'lock point' — the temporal position where variance curvature changes sign. R^2 measures fit quality.

Code	Level	Description & Trigger
S7-BWL-INF	INF	**HVLD bowl** HVLD vertex lock: bowl shape (a > 0). Variance is accelerating — system integrating. ***Trigger:*** *Emitted when step6_pll_shape == 'bowl'. Value: R^2 of the quadratic fit.*
S7-HIL-INF	INF	**HVLD hill** HVLD vertex lock: hill shape (a < 0). Variance is decelerating — system segregating. ***Trigger:*** *Emitted when step6_pll_shape != 'bowl' (i.e. 'hill'). Value: R^2 of the quadratic fit.*
S7-HRQ-INF	INF	**HVLD R^2 high** HVLD R^2 > 0.8. Quadratic fit describes the trajectory well. ***Trigger:*** *Emitted when step6_pll_R2 > 0.8. Indicates strong polynomial structure.*
S7-LRQ-WRN	WRN	**HVLD R^2 low** HVLD R^2 < 0.5. Quadratic fit is poor — trajectory may be non-polynomial.

		Trigger: *Emitted when step6_pll_R2 < 0.5. The quadratic model is a poor fit to the variance trajectory.*
S7-VTX-INF	**INF**	**Vertex location** HVLD vertex (lock point) location recorded. ***Trigger:*** *Always emitted. Value: step6_vertex (the temporal location of the quadratic vertex).*

3.6 S8 — Transcendental Squeeze

Step 8: Compares the HVLD vertex to a library of transcendental constants (e, π, 2π, e^π, etc.) and their reciprocals. Classifies the match as NATURAL (δ < 1%), INVESTIGATE (1–5%), or FLAG (> 5%). The squeeze is the pipeline's primary classification instrument.

Code	Level	Description & Trigger
S8-NAT-INF	**INF**	**NATURAL** HTP classification: NATURAL. At least one transcendental match within δ < 1%. ***Trigger:*** *Emitted when the closest transcendental constant match has delta < 0.01.*
S8-INV-WRN	**WRN**	**INVESTIGATE** HTP classification: INVESTIGATE. Nearest match δ between 1% and 5%. Consider alternate decomposition. ***Trigger:*** *Emitted when the closest match has 0.01 ≤ delta < 0.05.*
S8-FLG-WRN	**WRN**	**FLAG** HTP classification: FLAG. No match within 5%. System may be synthetic, adversarial, or improperly decomposed. ***Trigger:*** *Emitted when the closest match has delta ≥ 0.05, or when no squeeze result is available.*
S8-EUL-DIS	**DIS**	**Euler-family match** Nearest constant is from the Euler family (2π, e^π, π^e, or their reciprocals). Pipeline geometry resonates with system structure. ***Trigger:*** *Emitted when the matched constant name is in ['2pi', 'e^pi', 'pi^e', '1/(2pi)', '1/(e^pi)', '1/(pi^e)'].*
S8-SQZ-INF	**INF**	**Squeeze count** Number of transcendental proximity matches recorded. ***Trigger:*** *Always emitted. Value: step7_squeeze_count (how many constants fall within proximity).*
S8-TGT-DIS	**DIS**	**Tight match** Transcendental match tighter than δ < 0.001. Precision-level structural resonance. ***Trigger:*** *Emitted when the closest match has delta < 0.001. Indicates extremely precise alignment.*

3.7 S9 — EITT Entropy

Step 9: Computes Shannon entropy H(t) for all observations, normalised by H_max = ln(D). Tests EITT invariance: whether geometric-mean decimation preserves entropy within 5% at multiple compression ratios. Also runs chaos detection for stalls, spikes, and reversals in the angular velocity of the entropy trajectory.

Code	Level	Description & Trigger
S9-ENT-INF	**INF**	**Entropy computed** Shannon entropy H(t) computed for all observations. Normalised by H_max = ln(D). ***Trigger:*** *Always emitted. Value: step8_entropy_mean.*
S9-EIT-INF	**INF**	**EITT pass** EITT invariance test passed at all tested compression ratios (variation < 5%). ***Trigger:*** *Emitted when all entries in step8_eitt_invariance have pass == True.*

S9-EIF-WRN	WRN	**EITT fail** EITT invariance test failed at one or more compression ratios. ***Trigger:*** *Emitted when at least one entry in step8_eitt_invariance has pass == False.*
S9-CHS-INF	INF	**Chaos low** Chaos detection: few deviations. System is compositionally smooth. ***Trigger:*** *Emitted when step9_chaos_detection.total_deviations < 10.*
S9-CHH-DIS	DIS	**Chaos high** Chaos detection: many deviations. System exhibits compositional turbulence — stalls, spikes, or reversals. ***Trigger:*** *Emitted when step9_chaos_detection.total_deviations ≥ 10.*
S9-STL-DIS	DIS	**Stalls detected** Angular velocity stalls detected — entropy rate dropped below 15% of running mean. ***Trigger:*** *Emitted when step9_chaos_detection.stalls > 0.*
S9-SPK-DIS	DIS	**Spikes detected** Angular velocity spikes detected — entropy rate exceeded 3.5× running mean. ***Trigger:*** *Emitted when step9_chaos_detection.spikes > 0.*
S9-REV-DIS	DIS	**Reversals detected** Entropy reversals detected — ≥2 sign changes in 5 consecutive samples. ***Trigger:*** *Emitted when step9_chaos_detection.reversals > 0.*

3.8 SA — Ternary Projection

Step 10: Projects compositional data onto a 2D ternary diagram (D=3), a 1-simplex line (D=2), or a dimension-reduced projection (D>3). Provides visual confirmation of simplex geometry.

Code	Level	Description & Trigger
SA-TRN-INF	INF	**Ternary OK** Ternary projection computed. Type: ternary (D=3), 1-simplex (D=2), or projected (D>3). ***Trigger:*** *Always emitted after ternary projection completes. Unconditional.*

3.9 SB — Complex Plane

Step 11: Maps compositional data to centroid-relative complex plane coordinates. Enables angular analysis and phase detection in the composition space.

Code	Level	Description & Trigger
SB-CPX-INF	INF	**Complex plane OK** Complex plane mapping computed. Centroid-relative coordinates. ***Trigger:*** *Always emitted after complex plane mapping completes. Unconditional.*

3.10 SC — Helix/Polar

Step 12: Embeds compositional trajectories in 3D helix/polar coordinates. Computes radius, angle, and angular velocity for each observation, enabling temporal pattern detection.

Code	Level	Description & Trigger
SC-HLX-INF	INF	**Helix OK** 3D helix/polar embedding computed. Radius, angle, angular velocity. ***Trigger:*** *Always emitted after helix/polar embedding completes. Unconditional.*

3.11 XU — Extended Universal Panel

Extended: Always-on extended diagnostics: per-carrier contribution analysis, match density calibration, carrier drift detection, CoDa structural panel (variation matrix, CLR summary, Aitchison distance), claim-status summary, carrier hierarchy detection, conservation signature analysis, compositional fingerprint generation, and Component Power Mapper.

Code	Level	Description & Trigger
XU-PCC-INF	INF	**Carrier decomposition** Per-carrier contribution computed. Shows which carrier drives the variance. ***Trigger:*** *Emitted when extended.universal data is present. Value: per_carrier_contribution dict.*
XU-PCD-DIS	DIS	**Carrier dominance** One carrier contributes >60% of total variance. Strong structural signal. ***Trigger:*** *Emitted when per_carrier_contribution.dominant_pct > 60.*
XU-MDN-INF	INF	**Match density natural** Match density consistent with natural system (density >> synthetic baseline). ***Trigger:*** *Emitted when match_density.density > 10.*
XU-MDS-CAL	CAL	**Match density low** Match density below natural baseline. May indicate synthetic data or sparse constant coverage. ***Trigger:*** *Emitted when match_density.density ≤ 10.*
XU-DRI-INF	INF	**Drift stable** Carrier drift trend: stable. Second half compositionally similar to first half. ***Trigger:*** *Emitted when carrier_drift.drift_direction == 'stable'.*
XU-DRU-DIS	DIS	**Drift increasing** Carrier drift trend: increasing. System is evolving — second half more dispersed. ***Trigger:*** *Emitted when carrier_drift.drift_direction == 'increasing'.*
XU-DRD-DIS	DIS	**Drift decreasing** Carrier drift trend: decreasing. System is converging — second half more concentrated. ***Trigger:*** *Emitted when carrier_drift.drift_direction == 'decreasing'.*
XU-CDA-INF	INF	**CoDa panel OK** CoDa structural panel computed: variation matrix, CLR summary, Aitchison distance. ***Trigger:*** *Always emitted when extended.universal data is present.*
XU-CLM-INF	INF	**Claim summary** Claim-status and envelope summary generated. ***Trigger:*** *Always emitted when extended.universal data is present.*
XU-HRC-INF	INF	**Hierarchy moderate** Carrier hierarchy ratio (max/min of central values) is between 10^3 and 10^6. Moderate disparity — Dirichlet perturbation stable. ***Trigger:*** *Emitted when the max/min ratio of per-carrier contributions is between 10^3 and 10^6.*
XU-HRX-WRN	WRN	**Hierarchy extreme** Carrier hierarchy ratio exceeds 10^6 (but below Guard 4 limit of 10^{15}). Extreme disparity — perturbation methods may need alpha flooring. EITT failure is expected. ***Trigger:*** *Emitted when the max/min ratio of per-carrier contributions exceeds 10^6.*
XU-CSL-DIS	DIS	**Conservation signature** All carrier pairs show PID redundancy and at least 50% of ratio pairs are stable (CV < 20%). Strong signature of an underlying conservation law linking all carriers. ***Trigger:*** *Emitted when PID dominant mode is 'redundancy' AND at least 50% of ratio-pair lattice pairs have CV < 20%.*

XU-FPR-INF	**INF**	**Fingerprint generated** Compositional fingerprint computed: hash of HVLD shape, R^2 band, classification, structural modes, and nearest constant family. Use for cross-database matching. ***Trigger:*** *Emitted when fingerprint generation completes in the universal panel.*
XU-CPM-INF	**INF**	**Power map computed** Component Power Mapper analysis completed. Leverage index, phase boundary map, and power scores computed for all carriers. ***Trigger:*** *Emitted when power_map data is present in the result (from hs_sensitivity.py).*
XU-DPC-DIS	**DIS**	**Disproportionate carrier** One or more carriers have power-to-fraction ratios exceeding 2.0x, meaning their influence on system character is at least double what their mass fraction would predict. ***Trigger:*** *Emitted when power_scores.disproportionate_carriers is non-empty.*
XU-PSC-WRN	**WRN**	**Phase-sensitive carrier** One or more carriers have criticality margins below 0.3, meaning small changes to their values can flip the system classification or HVLD shape. ***Trigger:*** *Emitted when summary.phase_sensitive_carriers is non-empty.*
XU-CFR-WRN	**WRN**	**Classification flip risk** Perturbation of one or more carriers caused the system classification to flip between NATURAL and FLAG (or between shapes). ***Trigger:*** *Emitted when summary.classification_flip_risk_carriers is non-empty.*
XU-PWR-DIS	**DIS**	**Power ≠ mass** The power ranking of carriers differs from the mass fraction ranking. Component influence is not proportional to component fraction. This is the compositional leverage effect. ***Trigger:*** *Emitted when summary.rankings_differ is True.*

3.12 XC — Extended Conditional Panel

Extended: Conditionally-computed extended diagnostics (require D ≥ 3 or specific data properties): Partial Information Decomposition (PID), transfer entropy between carriers, order sensitivity testing, ratio-pair lattice analysis, and zero-crossing detection near the simplex boundary.

Code	Level	Description & Trigger
XC-PID-INF	**INF**	**PID computed** Partial Information Decomposition computed. Redundant, unique, and synergistic carrier interactions quantified. ***Trigger:*** *Emitted when conditional.PID data is present.*
XC-PIR-DIS	**DIS**	**PID redundancy** PID dominant mode: redundancy. Carriers share information — coupled dynamics. ***Trigger:*** *Emitted when PID.dominant == 'redundancy'.*
XC-PIU-DIS	**DIS**	**PID unique** PID dominant mode: unique. One carrier contributes information the others cannot — independent dynamics. ***Trigger:*** *Emitted when PID.dominant != 'redundancy' (i.e. 'unique' or 'synergistic').*
XC-TEE-INF	**INF**	**Transfer entropy** Transfer entropy matrix computed. Directed information flow between carriers quantified. ***Trigger:*** *Emitted when conditional.transfer_entropy data is present.*
XC-TEF-DIS	**DIS**	**TE dominant flow** Dominant directed information flow identified. One carrier informationally predicts another. ***Trigger:*** *Emitted when transfer_entropy.dominant_te > 0.001.*

XC-TEZ-CAL	**CAL**	**TE zero** Transfer entropy is zero across all pairs. System is compositionally memoryless at this aggregation level. ***Trigger:*** *Emitted when transfer_entropy.dominant_te ≤ 0.001.*
XC-ORD-INF	**INF**	**Order matters** Order sensitivity test: temporal/parametric order significantly affects the variance trajectory. ***Trigger:*** *Emitted when order_sensitivity.order_matters is True.*
XC-ORN-INF	**INF**	**Order neutral** Order sensitivity test: shuffling order does not change the result. Structure is cross-sectional. ***Trigger:*** *Emitted when order_sensitivity.order_matters is False.*
XC-RPL-INF	**INF**	**Ratio-pair lattice** Ratio-pair lattice computed. All pairwise log-ratios ranked by stability (CV). ***Trigger:*** *Emitted when conditional.ratio_pair_lattice data is present.*
XC-RPS-DIS	**DIS**	**Stable ratio found** A highly stable pairwise ratio detected (CV < 20%). These carriers are compositionally locked. ***Trigger:*** *Emitted for each ratio pair where CV < 20%. May fire multiple times.*
XC-RPV-DIS	**DIS**	**Volatile ratio** A highly volatile pairwise ratio detected (CV > 50%). These carriers are compositionally independent. ***Trigger:*** *Emitted for each ratio pair where CV > 50%. May fire multiple times.*
XC-ZCR-DIS	**DIS**	**Zero crossings** Carrier zero-crossing events detected. Compositions approaching simplex boundary — structural singularities. ***Trigger:*** *Emitted when conditional.zero_crossing data is present.*

3.13 RP — Report Summary

Post-pipeline: Final report codes confirming pipeline completion, determinism, and version. These always fire on a successful run.

Code	Level	Description & Trigger
RP-CMP-INF	**INF**	**Run complete** H^s extended pipeline run completed successfully. All results available for reporting. ***Trigger:*** *Always emitted at pipeline end.*
RP-DTM-INF	**INF**	**Deterministic** Pipeline is deterministic. Repeated runs on identical data produce bit-identical results. ***Trigger:*** *Always emitted at pipeline end.*
RP-VER-INF	**INF**	**Version** Pipeline version and configuration recorded. ***Trigger:*** *Always emitted. Value: pipeline_version string.*

4. Structural Mode Detection

Structural modes are second-order diagnoses. They are not individual measurements but geometric interpretations of the code pattern as a whole. The detection matrix scans all emitted codes and fires modes based on requires/forbids logic, evaluated in priority order. Each mode is an investigation prompt, not a verdict.

Detection algorithm: for each mode in priority order, check (1) ALL required code prefixes are present, (2) if requires_any is specified, at least one is present, (3) NO forbidden code prefixes are present. Special condition: SM-IND-DIS also requires ≥3 volatile ratio pairs and zero stable pairs.

SM-OVC-CAL Overconstrained (Priority 1)

Requires: S7-HRQ + S8-NAT + S8-TGT + S9-EIT

Forbids: S9-CHH, S8-FLG

System is highly constrained — near-perfect HVLD R^2, EITT pass, and precision-level δ. This is either a mathematical function, a physical system under tight constraints, or derived data. INVESTIGATE: verify this is measured data, not model output. If derived, the tight match is expected and calibrates the pipeline. If measured, this is a precision-standard-grade result.

SM-MCA-WRN Missing carrier (Priority 2)

Requires: S8-FLG + (XC-ZCR or S6-ZRO)

Forbids: None

Compositional geometry suggests an incomplete decomposition. FLAG classification combined with zero-crossings or degenerate vertices indicates energy or information is leaking to an untracked carrier. INVESTIGATE: what physical channel is absorbing the missing fraction? Add the missing carrier and re-run.

SM-BPO-DIS Bimodal population (Priority 3)

Requires: S7-HIL + S9-EIF + S9-CHH

Forbids: None

Two or more distinct compositional populations detected on the simplex. The variance trajectory shows mid-range peak (hill) with high turbulence and EITT failure — entropy decimation mixes populations that should be separate. INVESTIGATE: decompose by subpopulation.

SM-TNT-DIS Turbulent natural (Priority 4)

Requires: S8-NAT + S9-EIF + S9-CHH

Forbids: None

System is NATURAL but turbulent — transcendental match found despite EITT failure and high chaos. The turbulence is structural, not noise. INVESTIGATE: the turbulence pattern contains information. What mechanisms produce the oscillation?

SM-RTR-DIS Regime transition (Priority 5)

Requires: S9-STL + S9-REV + (XU-DRU or XU-DRD)

Forbids: None

System undergoes compositional regime change — drift detected with stalls and reversals. The composition is not stationary; it transitions between structural states. INVESTIGATE: the stall points mark regime boundaries.

SM-DGN-WRN Degenerate simplex (Priority 6)

Requires: XC-ZCR + (S8-FLG or S7-LRQ)

Forbids: None

Data approaches the simplex boundary — zero-crossings detected with FLAG or low R^2. One or more observations have a carrier at or near zero. INVESTIGATE: is the zero physical or artefactual?

SM-DMR-DIS Domain resonance (Priority 7)

Requires: S8-EUL + S8-NAT
Forbids: S8-FLG

System encodes deep geometric structure — Euler-family transcendental detected with strong R^2. The constant is not accidental; the pipeline's log-ratio geometry IS the structure. INVESTIGATE: compare this constant to other decompositions in the same domain.

SM-CPL-DIS Carrier coupling (Priority 8)

Requires: XC-PIR + XC-RPS
Forbids: None

Carriers are physically linked — PID shows redundancy and at least one ratio pair is highly stable. The coupled carriers share a common driver. INVESTIGATE: the stable ratio encodes a physical constraint.

SM-IND-DIS Carrier independence (Priority 9)

Requires: XC-RPV (3+ volatile pairs)
Forbids: XC-RPS, XC-PIR

Carriers are governed by independent mechanisms — all ratio pairs are volatile and PID shows unique information. No carrier predicts another. INVESTIGATE: decomposition may be maximally informative.

SM-SCG-INF Smooth convergence (Priority 10)

Requires: S7-BWL + S9-EIT
Forbids: S9-CHH, S8-FLG, S9-EIF

System converges smoothly to compositional equilibrium. Bowl HVLD with low turbulence and EITT preservation. This is the expected mode for well-sampled natural systems with a single dominant mechanism.

5. Code Generation and Reporting

5.1 generate_codes(result)

The generate_codes() function takes the complete pipeline result dictionary (from HigginsDecomposition.run_full_extended()) and walks through each stage's output, emitting codes based on threshold checks, existence checks, and classification outcomes. Each emitted code carries four fields: code (the SS-CCC-LLL identifier), short (a brief label), verbose (the full diagnostic description from CODE_DICTIONARY), and value (the measured quantity that triggered the code, where applicable).

The function processes stages in pipeline order: Guards → Closure → CLR → Variance → HVLD → Squeeze → EITT → Geometry → Extended Universal → Extended Conditional → Component Power → Report. After all individual codes are emitted, the function runs the structural mode detection pass, scanning the emitted code prefixes against the STRUCTURAL_MODES matrix.

5.2 codes_to_summary(codes)

The codes_to_summary() function takes the list of emitted code dicts and produces a human-readable report. It groups codes by severity level (ERR → WRN → DIS → CAL → INF), counts each category, and formats them for display. Errors show full verbose descriptions. Warnings and calibration signals show short labels. Discoveries show short labels plus their measured values. Information codes are counted but not individually listed (they form the structural skeleton).

5.3 Reading a Diagnostic Report

A complete diagnostic report from a healthy natural system typically shows 25–40 codes: a base of ~20 INF codes (one per step), 0 ERR codes, 0–2 WRN codes, 3–8 DIS codes (the scientific findings), and 0–1 CAL codes. The structural mode section at the end provides the geometric interpretation: SM-SCG (smooth convergence) for well-behaved systems, SM-DMR (domain resonance) for systems with deep structure, or SM-BPO/SM-RTR/SM-TNT for systems with complex dynamics.

A report with ERR codes indicates the pipeline rejected the data at the Guard stage. No further codes will be emitted. A report with FLAG (S8-FLG-WRN) and multiple warnings suggests the decomposition may need revision: check carrier definitions, add missing carriers, or split heterogeneous data.

6. Quick Reference: All 76 Codes

Code	Lvl	Short Name	Description
GD — Guard (Input Validation)			
GD-D2M-ERR	ERR	D < 2	Input rejected: D must be >= 2. The simplex requires at least 2 compositional parts.
GD-N5M-ERR	ERR	N < 5	Input rejected: N must be >= 5. The variance trajectory needs at least 3 non-zero points for a meaningful HVLD fit.
GD-NAN-ERR	ERR	NaN detected	Input rejected: data contains NaN values. Clean or impute before decomposition.
GD-INF-ERR	ERR	Inf detected	Input rejected: data contains Inf values. Check for division by zero or overflow.
GD-SCL-ERR	ERR	Scale disparity	Input rejected: max/min ratio exceeds 1e15. Normalise or partition before decomposition.
GD-ALL-INF	INF	All guards pass	All 4 input guards passed. Data is valid for H^s pipeline processing.
S4 — Simplex Closure			
S4-CLO-INF	INF	Closure OK	Simplex closure successful. All rows sum to 1.0 within tolerance 1e-12.
S4-ZRP-INF	INF	Zero replacement	Multiplicative zero replacement applied. ZERO_DELTA = 1e-6.
S4-ZRN-INF	INF	No zeros	No zero replacement needed. All input values are positive.
S5 — CLR Transform			
S5-CLR-INF	INF	CLR complete	Centred log-ratio transform applied. All rows sum to zero in CLR space.
S6 — Aitchison Variance			
S6-VAR-INF	INF	Variance computed	Cumulative Aitchison variance trajectory σ^2_A(t) computed for all time indices.
S6-ZRO-WRN	WRN	Zero variance	σ^2_A is zero or near-zero. Data may be constant or degenerate.
S6-RNG-INF	INF	Variance range	Variance trajectory range and final value recorded.
S7 — HVLD Vertex Lock			
S7-BWL-INF	INF	HVLD bowl	HVLD vertex lock: bowl shape (a > 0). Variance is accelerating — system integrating.
S7-HIL-INF	INF	HVLD hill	HVLD vertex lock: hill shape (a < 0). Variance is decelerating — system segregating.
S7-HRQ-INF	INF	HVLD R^2 high	HVLD R^2 > 0.8. Quadratic fit describes the trajectory well.
S7-LRQ-WRN	WRN	HVLD R^2 low	HVLD R^2 < 0.5. Quadratic fit is poor — trajectory may be non-polynomial.
S7-VTX-INF	INF	Vertex location	HVLD vertex (lock point) location recorded.
S8 — Transcendental Squeeze			
S8-NAT-INF	INF	NATURAL	HTP classification: NATURAL. At least one transcendental match within δ < 1%.
S8-INV-WRN	WRN	INVESTIGATE	HTP classification: INVESTIGATE. Nearest match δ between 1% and 5%. Consider alternate decomposition.
S8-FLG-WRN	WRN	FLAG	HTP classification: FLAG. No match within 5%. System may be synthetic, adversarial, or improperly decomposed.
S8-EUL-DIS	DIS	Euler-family match	Nearest constant is from the Euler family (2π, e^π, π^e, or their reciprocals). Pipeline geometry resonates with system ...
S8-SQZ-INF	INF	Squeeze count	Number of transcendental proximity matches recorded.
S8-TGT-DIS	DIS	Tight match	Transcendental match tighter than δ < 0.001. Precision-level structural resonance.
S9 — EITT Entropy			
S9-ENT-INF	INF	Entropy computed	Shannon entropy H(t) computed for all observations. Normalised by H_max = ln(D).
S9-EIT-INF	INF	EITT pass	EITT invariance test passed at all tested compression ratios (variation < 5%).
S9-EIF-WRN	WRN	EITT fail	EITT invariance test failed at one or more compression ratios.
S9-CHS-INF	INF	Chaos low	Chaos detection: few deviations. System is compositionally smooth.
S9-CHH-DIS	DIS	Chaos high	Chaos detection: many deviations. System exhibits compositional turbulence — stalls, spikes, or reversals.
S9-STL-DIS	DIS	Stalls detected	Angular velocity stalls detected — entropy rate dropped below 15% of running mean.
S9-SPK-DIS	DIS	Spikes detected	Angular velocity spikes detected — entropy rate exceeded 3.5× running mean.

S9-REV-DIS	DIS	Reversals detected	Entropy reversals detected — ≥2 sign changes in 5 consecutive samples.
SA — Ternary Projection			
SA-TRN-INF	INF	Ternary OK	Ternary projection computed. Type: ternary (D=3), 1-simplex (D=2), or projected (D>3).
SB — Complex Plane			
SB-CPX-INF	INF	Complex plane OK	Complex plane mapping computed. Centroid-relative coordinates.
SC — Helix/Polar			
SC-HLX-INF	INF	Helix OK	3D helix/polar embedding computed. Radius, angle, angular velocity.
XU — Extended Universal Panel			
XU-PCC-INF	INF	Carrier decomposition	Per-carrier contribution computed. Shows which carrier drives the variance.
XU-PCD-DIS	DIS	Carrier dominance	One carrier contributes >60% of total variance. Strong structural signal.
XU-MDN-INF	INF	Match density natural	Match density consistent with natural system (density >> synthetic baseline).
XU-MDS-CAL	CAL	Match density low	Match density below natural baseline. May indicate synthetic data or sparse constant coverage.
XU-DRI-INF	INF	Drift stable	Carrier drift trend: stable. Second half compositionally similar to first half.
XU-DRU-DIS	DIS	Drift increasing	Carrier drift trend: increasing. System is evolving — second half more dispersed.
XU-DRD-DIS	DIS	Drift decreasing	Carrier drift trend: decreasing. System is converging — second half more concentrated.
XU-CDA-INF	INF	CoDa panel OK	CoDa structural panel computed: variation matrix, CLR summary, Aitchison distance.
XU-CLM-INF	INF	Claim summary	Claim-status and envelope summary generated.
XU-HRC-INF	INF	Hierarchy moderate	Carrier hierarchy ratio (max/min of central values) is between 10^3 and 10^6. Moderate disparity — Dirichlet perturbation ...
XU-HRX-WRN	WRN	Hierarchy extreme	Carrier hierarchy ratio exceeds 10^6 (but below Guard 4 limit of 10^{15}). Extreme disparity — perturbation methods may need...
XU-CSL-DIS	DIS	Conservation signature	All carrier pairs show PID redundancy and at least 50% of ratio pairs are stable (CV < 20%). Strong signature of an unde...
XU-FPR-INF	INF	Fingerprint generated	Compositional fingerprint computed: hash of HVLD shape, R^2 band, classification, structural modes, and nearest constant ...
XU-CPM-INF	INF	Power map computed	Component Power Mapper analysis completed. Leverage index, phase boundary map, and power scores computed for all carrier...
XU-DPC-DIS	DIS	Disproportionate carrier	One or more carriers have power-to-fraction ratios exceeding 2.0x, meaning their influence on system character is at lea...
XU-PSC-WRN	WRN	Phase-sensitive carrier	One or more carriers have criticality margins below 0.3, meaning small changes to their values can flip the system class...
XU-CFR-WRN	WRN	Classification flip risk	Perturbation of one or more carriers caused the system classification to flip between NATURAL and FLAG (or between shape...
XU-PWR-DIS	DIS	Power ≠ mass	The power ranking of carriers differs from the mass fraction ranking. Component influence is not proportional to compone...
XC — Extended Conditional Panel			
XC-PID-INF	INF	PID computed	Partial Information Decomposition computed. Redundant, unique, and synergistic carrier interactions quantified.
XC-PIR-DIS	DIS	PID redundancy	PID dominant mode: redundancy. Carriers share information — coupled dynamics.
XC-PIU-DIS	DIS	PID unique	PID dominant mode: unique. One carrier contributes information the others cannot — independent dynamics.
XC-TEE-INF	INF	Transfer entropy	Transfer entropy matrix computed. Directed information flow between carriers quantified.
XC-TEF-DIS	DIS	TE dominant flow	Dominant directed information flow identified. One carrier informationally predicts another.
XC-TEZ-CAL	CAL	TE zero	Transfer entropy is zero across all pairs. System is compositionally memoryless at this aggregation level.
XC-ORD-INF	INF	Order matters	Order sensitivity test: temporal/parametric order significantly affects the variance trajectory.
XC-ORN-INF	INF	Order neutral	Order sensitivity test: shuffling order does not change the result. Structure is cross-sectional.
XC-RPL-INF	INF	Ratio-pair lattice	Ratio-pair lattice computed. All pairwise log-ratios ranked by stability (CV).
XC-RPS-DIS	DIS	Stable ratio found	A highly stable pairwise ratio detected (CV < 20%). These carriers are compositionally locked.

XC-RPV-DIS	**DIS**	Volatile ratio	A highly volatile pairwise ratio detected (CV > 50%). These carriers are compositionally independent.
XC-ZCR-DIS	**DIS**	Zero crossings	Carrier zero-crossing events detected. Compositions approaching simplex boundary — structural singularities.
RP — Report Summary			
RP-CMP-INF	**INF**	Run complete	H^s extended pipeline run completed successfully. All results available for reporting.
RP-DTM-INF	**INF**	Deterministic	Pipeline is deterministic. Repeated runs on identical data produce bit-identical results.
RP-VER-INF	**INF**	Version	Pipeline version and configuration recorded.